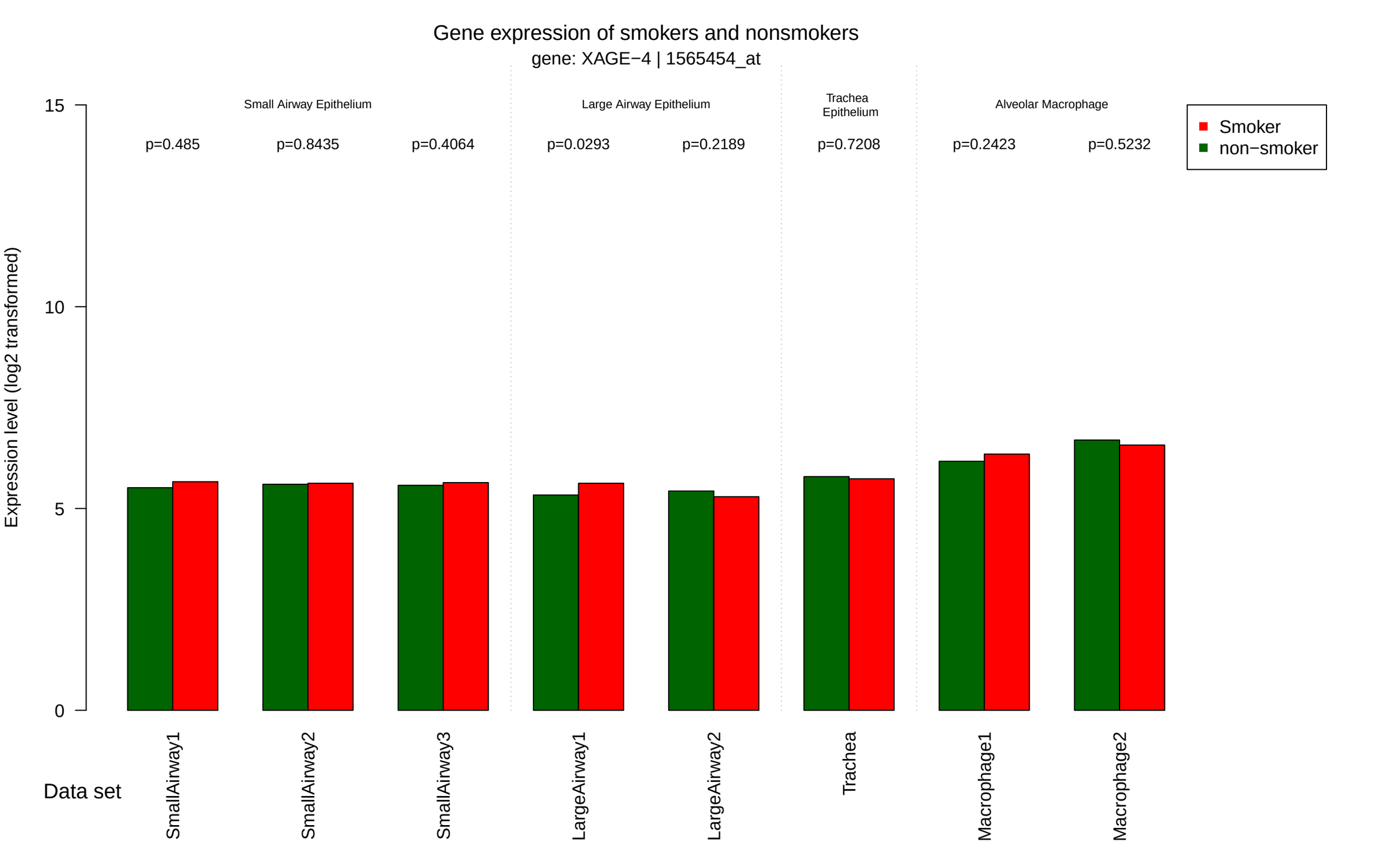
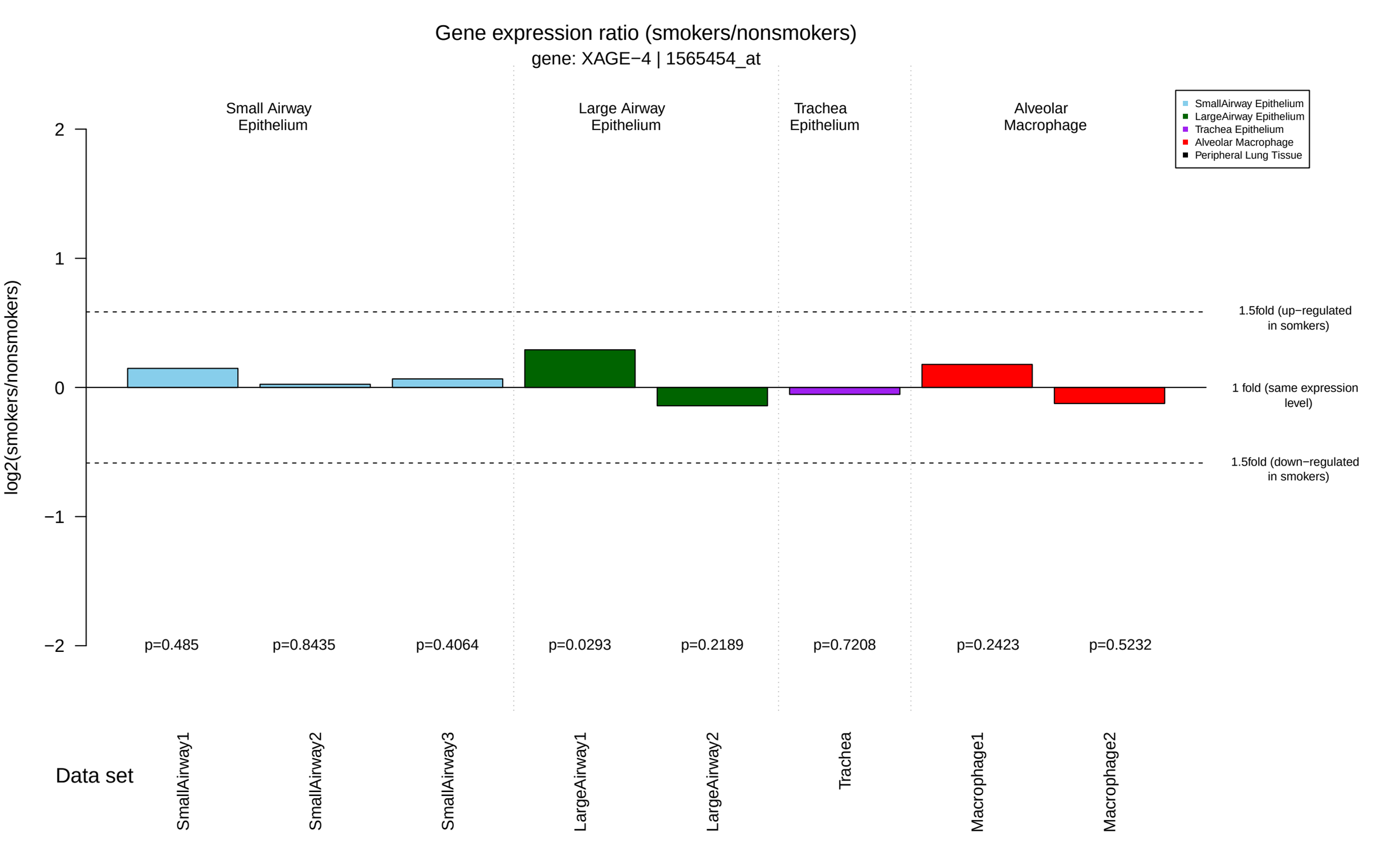
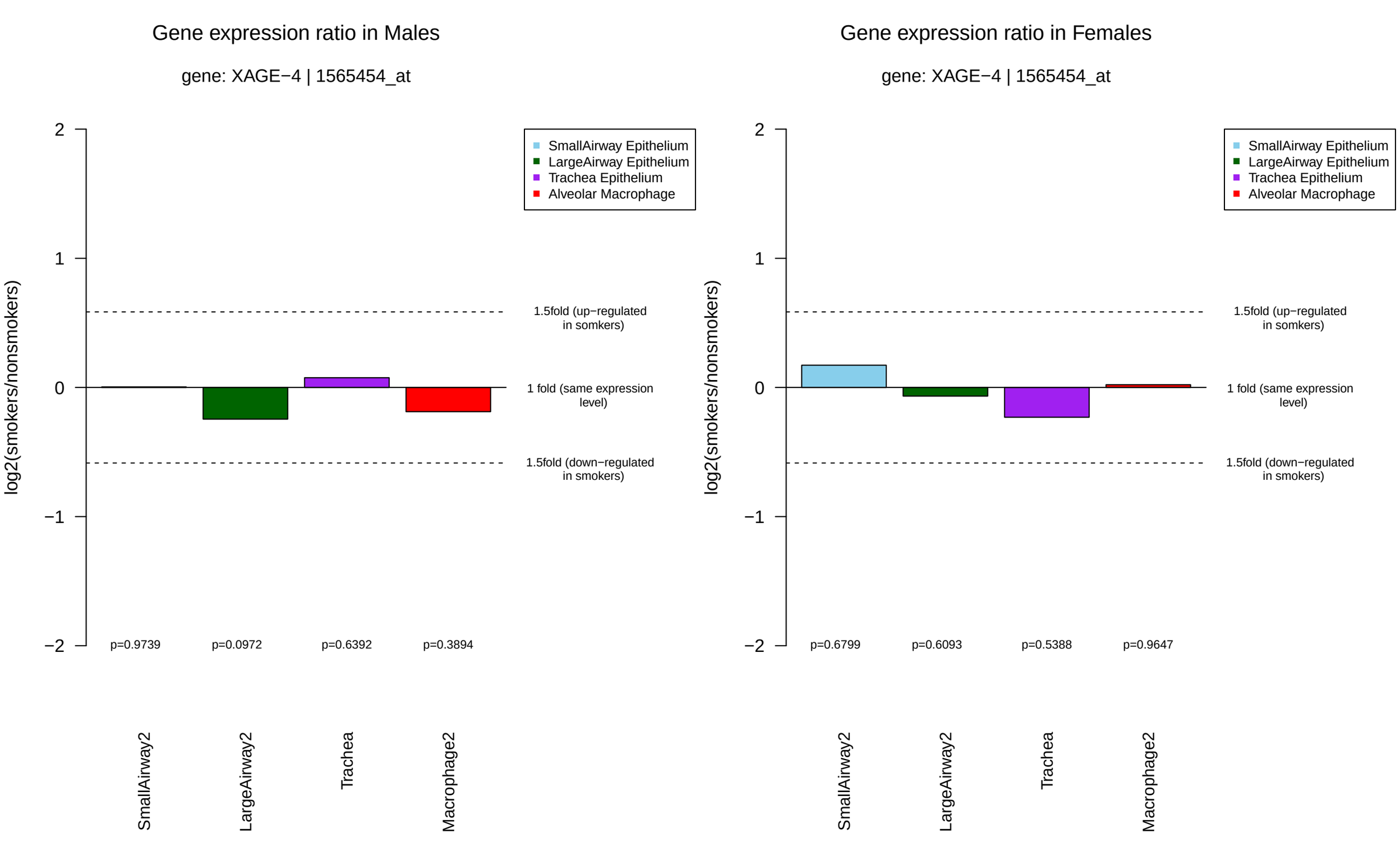
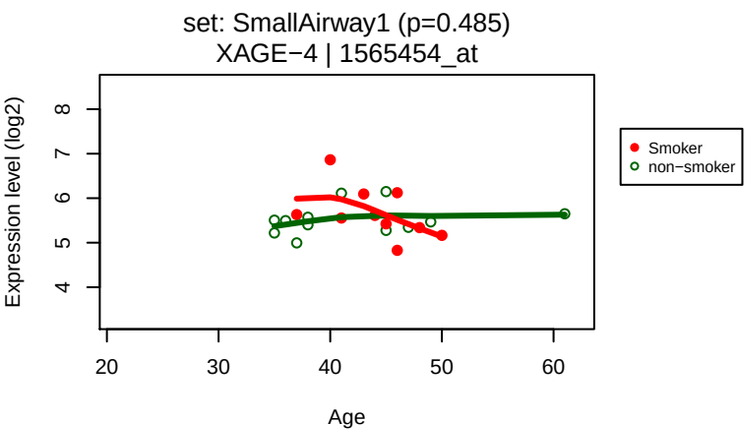
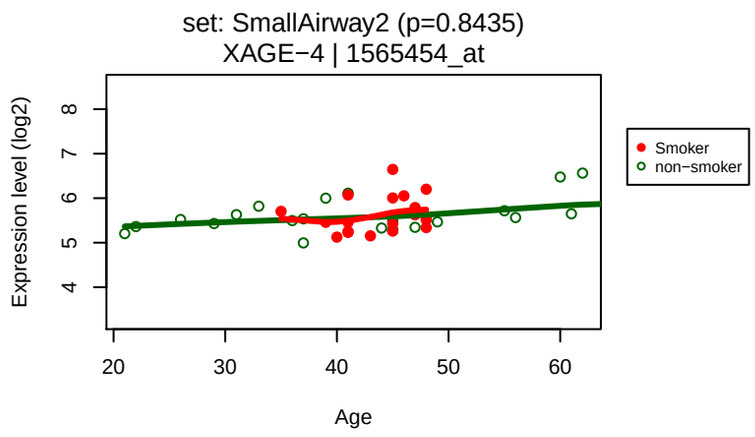
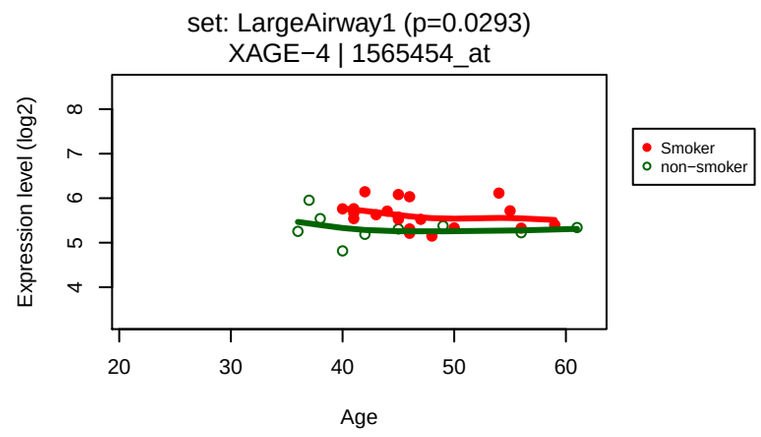

Gene expression of smokers and nonsmokers
gene: XAGE−4 | 1565454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.485
p=0.8435
p=0.4064
p=0.0293
p=0.2189
p=0.7208
p=0.2423
p=0.5232
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

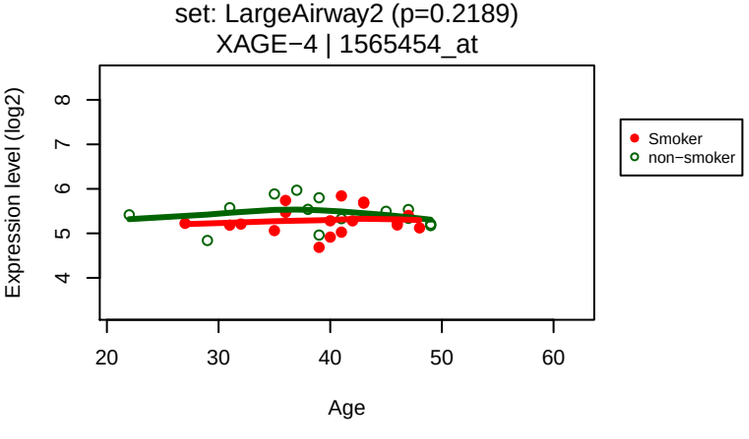

Gene expression ratio (smokers/nonsmokers)
gene: XAGE−4 | 1565454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.485
p=0.8435
p=0.4064
p=0.0293
p=0.2189
p=0.7208
p=0.2423
p=0.5232
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

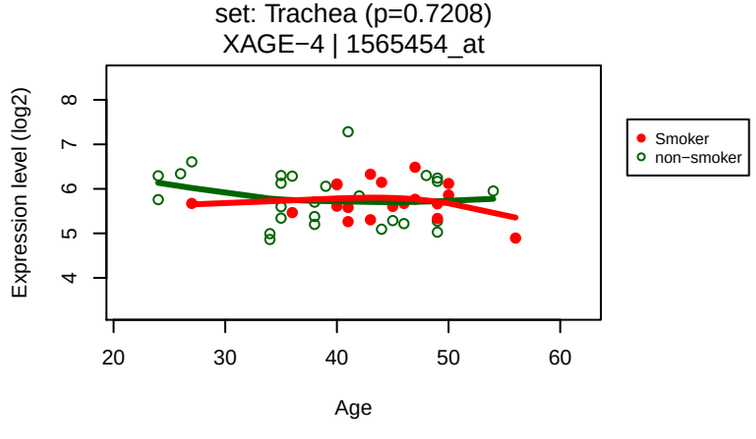

Gene expression ratio in Males
gene: XAGE−4 | 1565454_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9739
p=0.0972
p=0.6392
p=0.3894
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XAGE−4 | 1565454_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6799
p=0.6093
p=0.5388
p=0.9647
SmallAirway2
LargeAirway2
Trachea
Macrophage2

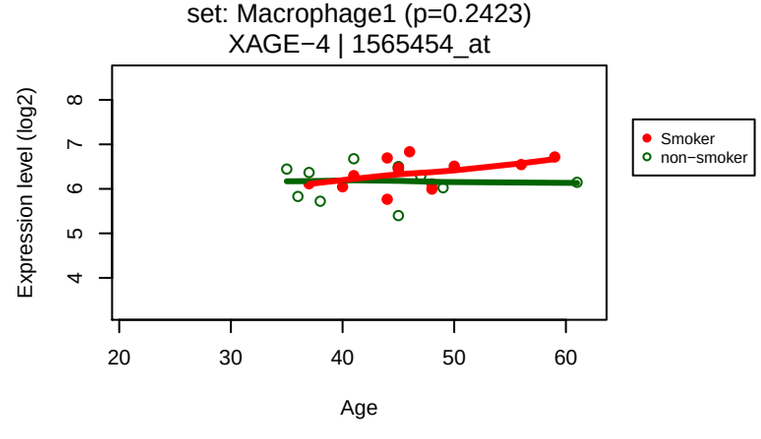

set: SmallAirway1 (p=0.485)
XAGE−4 | 1565454_at
Expression level (log2)
Age
Smoker
non−smoker

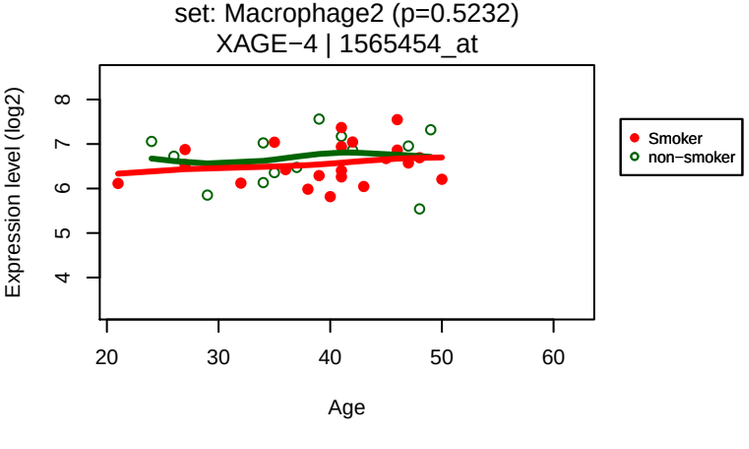

set: SmallAirway2 (p=0.8435)
XAGE−4 | 1565454_at
Expression level (log2)
Age
Smoker
non−smoker

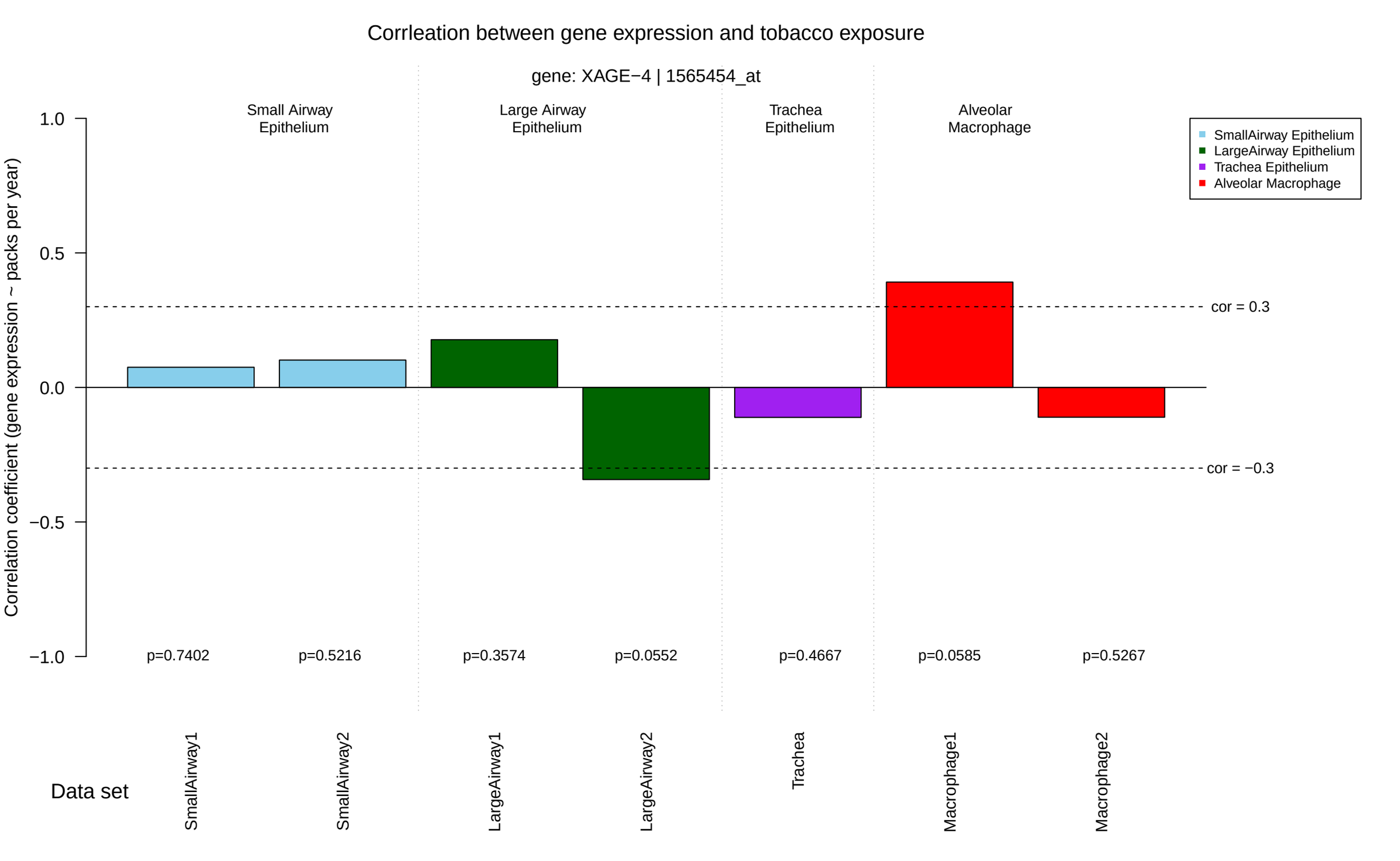

set: LargeAirway1 (p=0.0293)
XAGE−4 | 1565454_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2189)
XAGE−4 | 1565454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7208)
XAGE−4 | 1565454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2423)
XAGE−4 | 1565454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5232)
XAGE−4 | 1565454_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XAGE−4 | 1565454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7402
p=0.5216
p=0.3574
p=0.0552
p=0.4667
p=0.0585
p=0.5267
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2